

TIC 306294675

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	2.982219 d	VSX	no match
Epoch (t0)	1570.3514 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	2.434 % (24,341 ppm)	Gaia variable classification	no match
SNR	179.8	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	14 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	1.00 (gate >= 0.50)		
Robust depth sigma	81.4 (gate >= 7.0)		
Odd/even z-score	0.15 (consistent)		
Secondary eclipse	detected, phase -0.43, 5.6 sigma		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 86.0 sigma t0 refined by -0.0050 phase

Difference-image peak SNR 24.67 (gate >= 5.0) Compactness 0.97 (gate >= 0.50)

Dimming centroid offset 0.31 px (7 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 306294642

0.54 px away, dTmag +2.96 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

